Lifetime Asset Allocation in New Zealand: A Simulation and Optimization Approach

Callum Davidson

May 4, 2026

1 Foreword and Research Context

This research is conducted in partnership with Consilium NZ LTD (hereafter, Consilium). Consilium is a New Zealand-based financial services company that provides tools and consulting services to financial advisers and their clients in areas including investment management, portfolio construction, risk management, and financial planning. Consilium also offers managed funds and portfolio solutions designed to support advisers in meeting a wide range of client needs. The aim of this research partnership is to examine and improve lifetime asset allocation strategies for individual investors, with particular focus on retirees in the New Zealand context. Our objective is to produce insights and recommendations that may help Consilium better support advisers and their clients.

Although this research is conducted in partnership with Consilium, the views expressed in this paper are ours and do not necessarily reflect the official policy or position of Consilium. We retain full academic independence in the design, analysis, and reporting of this work. We also retain full ownership of the intellectual property associated with this research, while granting Consilium a non-exclusive licence to use the findings internally and share them with clients and partners. No confidential or proprietary information provided by Consilium has been used in this study.

We thank Consilium for their support and collaboration in this research project, and for the feedback that has helped shape the direction and focus of this work.

Contents

1 Foreword and Research Context **1**

2 Introduction **2**
- 2.1 The Status Quo on Lifetime Asset Allocation . . . 3

3 Retirement in New Zealand **5**
- 3.1 Spending By Age in New Zealand . . . 5
- 3.2 Bequests and Precautionary Savings . . . 6
- 3.3 Spending by Household Composition and Wealth . . . 7

4 Modelling Perceptions of Risk and Utility **7**

5 Methods for Simulating Lifetime Wealth Trajectories **7**
- 5.1 Modelling Dynamic Investor Behaviour . . . 8
  - 5.1.1 Wealth-Responsive Spending Policy . . . 8
  - 5.1.2 Wealth-Responsive Allocation Policy . . . 9
- 5.2 Methods for Simulating Asset Returns . . . 10

6 Methods for Optimising Asset Allocation Strategies **10**

7 Optimal Lifetime Asset Allocation Strategies for New Zealand Retirees **10**
- 7.1 Optimal Age-Based Glide-Paths . . . 11
- 7.2 Optimal Wealth-Responsive Allocation Strategies . . . 11

8 Discussion and Implications **11**

9 Conclusion **11**

A Replication and Discussion of Control Matrix Optimisation as Proposed by Mäkinen and Toivanen (2024) **14**

2 Introduction

Managing a retiree's nest egg requires navigating two broad forms of financial risk. A well-established principle in investment research is that assets with higher expected returns tend to come with higher volatility (e.g., (Campbell, 1996; Ghysels et al., 2005; Lundblad, 2007)). Although many investors spend considerable energy selecting particular securities or funds, Ibbotson and Kaplan (2001) find that roughly 90% of the variation in fund returns is explained by the allocation to major asset classes such as equities, bonds, and cash. As a result, asset allocation is the primary driver of a portfolio's long-term behaviour. For the typical investor, whose retirement savings are usually held in managed funds that do not allow meaningful security selection, the most important strategic decision is choosing the overall exposure to these broad asset classes.

The central challenge for individuals approaching retirement is finding an allocation between safe and risky assets that provides the returns required for a comfortable lifestyle, while avoiding excessive exposure to loss. Achieving this balance is not straightforward. Once the allocation is considered as a function of risk appetite, savings capacity, spending needs, lifespan, and total accumulated wealth, the problem becomes complex. Many of these inputs differ significantly between individuals, and even small differences in behaviour or circumstance can lead to materially different outcomes. The mathematical problem of finding an optimal allocation through time is therefore highly sensitive to real-world considerations. This complexity has contributed to the emergence of various heuristics and rules of thumb that attempt to provide simple guidance in the absence of a fully individualised solution.

The significance of this problem extends beyond individual welfare. The global asset-backed pension market was valued at 68.9 trillion USD in 2024 (OECD, 2025). For many people, their retirement savings represent the largest financial asset they will ever manage apart from their home. Consequently, the question of how to allocate this wealth over the course of a lifetime is significant not only for individuals but also for governments. Even modest improvements in the efficiency of retirement strategies can reduce pressure on public pension systems. In New Zealand, for example, every resident aged 65 or older receives New Zealand Superannuation regardless of income or assets. The programme is funded through general taxation. Due to an ageing population and ongoing increases in life expectancy, the long-term fiscal sustainability of this system is becoming increasingly challenging. A recent Treasury report (The Treasury, 2025) estimates that spending on New Zealand Superannuation and healthcare could account for nearly half of all government expenditure by 2065 if current policy settings remain unchanged. This has led to suggestions that New Zealanders will need to save more for retirement, for example through higher KiwiSaver contributions, and that adjustments to eligibility age or means testing may be required. There is already a broad consensus that the public is under-saving for retirement (e.g., (Dexter, 2025; Gilbert, 2025; Luxon, 2025; Palmer, 2025)).

A critical requirement for the success of such policies is ensuring that individuals make appropriate decisions about their lifetime asset allocation. At present, the New Zealand government provides little guidance on how individuals should allocate their retirement savings. The Financial Markets Authority requires that default KiwiSaver funds adopt a Balanced risk profile (Financial Markets Authority, 2023), defined as holding between 35 percent and 63 percent growth assets (Financial Markets Authority, 2024). This replaced the previous Conservative default in 2021 (Retirement Commission Te Ara Ahunga Ora, n.d.), after it became clear that many KiwiSaver members did not switch funds from the default fund and that a conservative allocation was poorly aligned with the needs of long-term investors (Ministry of Business, Innovation and Employment and The Treasury, 2019). Although a Balanced fund is a reasonable one-size-fits-most solution, it is likely too conservative for many younger investors who have long investment horizons and the capacity to tolerate short-term volatility.

This paper has three objectives. First, we review the current literature on lifetime asset allocation, with particular attention to the glide-path strategies that dominate both academic discussions and industry practice. We also examine the different risk measures and optimisation frameworks that researchers have used to evaluate such strategies. Second, we analyse the spending needs and risk concerns of typical retirees, with a particular focus on the New Zealand context. Understanding these patterns is essential for designing realistic simulations of retirement behaviour and for evaluating the amount of risk that individuals can tolerate. Finally, we develop and test a flexible simulation and optimisation framework for lifetime asset allocation. Building on the work of Mäkinen and Toivanen (2024), who propose a Monte Carlo portfolio optimisation framework, we adapt and extend their method to address the specific challenges that arise in lifetime allocation problems. We also modify their risk measure so that it reflects the practical concerns of real-world investors.

Our aim is to show that this framework can generate stable and robust strategies, while remaining flexible enough to incorporate a variety of risk measures, asset return models, and spending assumptions. In doing so, we aim both to test the validity of existing glide-path recommendations and to provide a foundation for further research in this area of personal finance.

2.1 The Status Quo on Lifetime Asset Allocation

In order to understand the rationale for any allocation strategy, we first require a clear picture of the financial life-cycle faced by a typical investor. Broadly, this can be divided into two phases: accumulation and decumulation. During accumulation, which typically spans early adulthood to retirement, investors build wealth through regular contributions and investment returns. Decumulation begins at retirement and continues until death; during this phase, investors withdraw funds to finance living expenses and other needs. This framing implies a life-cycle wealth trajectory in which wealth accumulates during working life, peaks around retirement, and may decline thereafter. Although this trajectory is often assumed to be hump-shaped, the extent to which real-world investor wealth actually declines in retirement (or whether it typically declines at all) remains an open question, to which we return later.

Another important concept in this discussion is human capital, which refers to the present value of an individual's future earnings potential. During the accumulation phase, human capital is typically high, as the investor has many working years ahead. As the investor approaches retirement, human capital declines, and at retirement, it effectively becomes zero. This shift in human capital has important implications for asset allocation decisions, as it affects the investor's overall risk profile and capacity to absorb investment losses.

The sheer amount of complexity in identifying optimal life-cycle allocation strategies is substantial. The optimal strategy depends on many factors and, likely because of this complexity, a large number of rules of thumb and financial products have emerged to guide investors and advisers. Most of these rules recommend a "declining-risk" strategy, in which exposure to equities decreases with age. This idea has been influential for decades and was even considered as part of potential reforms to the United States Social Security system under the George W. Bush administration (R. Shiller, 2005; R. J. Shiller, 2006). The fact that it was considered for such a large and consequential public pension system highlights the appeal, to both policymakers and governments, of improving retirement outcomes through better asset allocation.

A well-known example of this type of recommendation is the family of "age in bonds" rules, which suggest holding a bond allocation equal (or proportional) to the investor's age. The implied life-cycle policy is approximately linearly declining in risk exposure: high equity weight at younger ages, decreasing over time, hence "declining-risk" or "declining-equity". This approach appears in investment textbooks (e.g., (Bengen, 1996; Bodie et al., 2023)) and in popular advice literature (Choi, 2022). Dolvin et al. (2010) review U.S. target-date funds and find that most follow some form of declining-risk structure, often resembling a 120-minus-age equity rule. In New Zealand, SuperLife's Age Steps product follows a similar pattern, reducing equity exposure from 99% at age step 18 to 10% at age step 80 (see Figure 2.1).

The intuitive logic behind declining-risk strategies is that older investors have less time to recover from losses, and that their spending needs become more immediate. In this view, reducing exposure to risky assets as retirement approaches is considered prudent. However, continuing to reduce risk exposure throughout retirement may not be optimal, and academic support for this approach is mixed at best; much of the evidence instead indicates that constant or even increasing-risk strategies may be more effective for retirees undergoing decumulation.

The first mathematically rigorous treatment, by Merton (1969), showed that under lifetime consumption-utility maximisation, the optimal allocation is constant through time and does not depend on age. However, that conclusion relies on an investor withdrawing a fixed proportion of wealth each period (rather than a fixed dollar amount) and on asset returns being independent and identically distributed (IID) through time. Even so, using empirical retirement data, R. Shiller (2005) and R. J. Shiller (2006) find that declining-risk strategies may be inferior to a constant 100 percent equity allocation for retirees making fixed-dollar withdrawals. Moreover, Estrada (2014, 2015) find that declining-risk glide-paths tend to underperform both constant-risk and increasing-risk strategies across a wide range of risk measures, asset classes, and investor types. Using bootstrapped historical return data, they show that declining-risk strategies produce lower terminal wealth without meaningful downside-risk improvement.

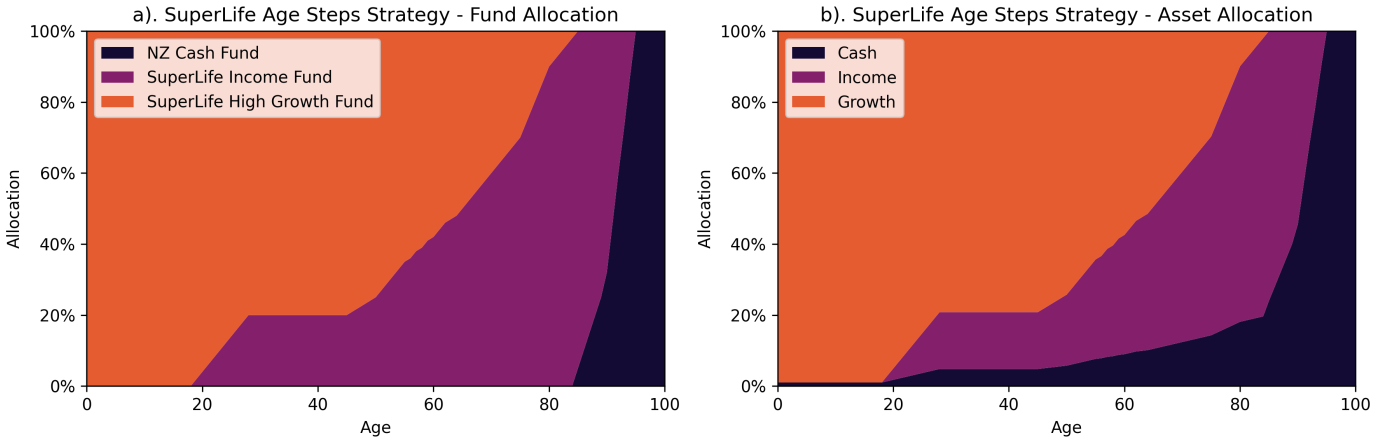

Figure 1: SuperLife's Age Steps target strategy as per their Statement of Investment Policy and Objectives (SIPO), showing the allocation to different funds (a) and the resulting allocation to different asset classes (b). Source: SuperLife (2026).

Although this paper will be principally concerned with the decumulation phase, it is worth noting that the evidence for declining-risk strategies during the accumulation phase is also mixed. The most robust theoretical support for declining-risk strategies during the accumulation phase comes from Bodie et al. (1992) and Cocco et al. (2005), who show that when human capital is taken into account, younger investors should hold a higher proportion of equities. In their models, human capital is treated as a bond-like asset because it provides a steady stream of income that is less volatile than equity returns. As investors age and their human capital declines, their overall risk exposure increases, which justifies a reduction in equity holdings. In this model, they essentially argue that the declining-risk strategy is appropriate during the accumulation phase, insofar as it offsets the changing risk profile due to declining human capital.

However, empirical evidence suggests that even during accumulation, declining-risk strategies may not be optimal. Anarkulova et al. (2025) find that for typical young investors with low wealth and savings rates, a constant 100 percent equity allocation provides similar downside protection to declining-risk strategies while delivering substantially higher expected returns. Using block-bootstrap resampled data from U.S. and international markets, they show that optimised glide-paths still underperform all-equity portfolios, except in the very earliest retirement years. They argue that fixed-income assets provide limited benefit once time-dependent return dynamics are incorporated, and that earlier studies tended to favour bonds because they assumed IID returns. In their optimal glide-paths, bonds appear only modestly, mostly at retirement onset, with allocations typically around 30–40 percent or less. This produces a U-shaped risk trajectory in which risky allocations dip at retirement and then rise.

This U-shaped risk trajectory is a common finding in the literature on optimal glide-paths, and it is often interpreted as evidence that some reduction in risk exposure at retirement is optimal. It appears again, more strongly, in work by Estrada (2015), who find that optimal glide-paths often involve a significant reduction in equity exposure at retirement, followed by a gradual increase in equity exposure later in retirement. This pattern is consistent with the idea that retirees are most vulnerable to sequence-of-returns risk at the start of retirement, when they have high accumulated wealth and low human capital. By holding some bonds at this point, retirees can reduce the likelihood of depleting their portfolio too quickly. However, as withdrawals reduce total wealth and the investor is further along in retirement, the impact of a run of bad returns becomes less severe (in real-dollar terms), and therefore a higher equity allocation becomes more acceptable in fixed-spending scenarios.

Although declining-risk strategies remain widely recommended in practice, the academic and mathematical literature does not strongly support their use during decumulation. Instead, the evidence generally suggests that constant or even increasing-risk strategies may produce better outcomes. While decreasing risk exposure may be appropriate near retirement during late accumulation, given heightened sequence-of-returns risk, the evidence suggests that once decumulation begins retirees may be better served by maintaining or even increasing risky-asset exposure.

The benefits of strategies such as SuperLife's Age Steps are therefore likely to be limited, and may even be counterproductive for many retirees. The broader evidence suggests that common guidance on life-cycle allocation may be incomplete, and that advisers may have scope to differentiate through more evidence-based portfolio design. It should be noted, however, that the literature is not fully settled: conclusions still depend on the risk measure used, return-process assumptions, and the treatment of spending and withdrawals. Many existing models also remain limited in their representation of real-world investor behaviour and market dynamics.

Our goal will be to study and model investor concerns and behaviour in a way that is more realistic and relevant to the New Zealand context. We will then use this understanding to inform the design of a flexible simulation and optimisation framework for lifetime asset allocation, which can be used to test the validity of existing glide-path recommendations and to explore alternative strategies that may better meet the needs of New Zealand retirees.

Implications for Consilium NZ LTD

While Consilium does not currently offer a glide-path product, nor explicitly recommend a declining-risk strategy, the existing literature still has clear implications for their business. It suggests that prevailing views on life-cycle allocation may be incomplete, and that there may be scope for Consilium to differentiate through more sophisticated, evidence-based portfolio advice. In particular, there may be an opportunity to challenge adviser assumptions regarding the need for declining-risk strategies.

Consilium has typically based much of its modelling of portfolio performance on IID return assumptions, which may not accurately reflect real-world market dynamics. By investigating asset allocation strategies that account for more realistic return dynamics, Consilium can provide more robust and effective advice to their clients.

3 Retirement in New Zealand

The New Zealand-specific literature on retirement concerns remains limited, so we draw extensively on international evidence. Understanding the institutional context is essential for interpreting this evidence appropriately. New Zealand's retirement system combines a universal public pension (New Zealand Superannuation) with voluntary private savings through KiwiSaver. New Zealand Superannuation provides all residents aged 65 and over with a baseline income regardless of work history or assets, and adjusts annually for inflation. KiwiSaver allows voluntary contributions from employees, matched by employers and the government. Together, these mechanisms provide a safety net while encouraging personal saving (Ministry of Social Development, 2022; Te Ara Ahunga Ora: Retirement Commission, 2021).

KiwiSaver was introduced in 2007, so most current retirees have little or no accumulated balance. Currently, 40 percent of New Zealanders aged 65 rely entirely on New Zealand Superannuation, with a further 20 percent holding minimal additional savings (Retirement Commission Te Ara Ahunga Ora, 2022). As the scheme matures and government policy encourages higher contribution rates (Luxon, 2025), future cohorts will retire with substantially larger balances. This demographic shift increases the need for robust guidance on asset allocation and withdrawal strategies during retirement.

Given these trends, the Australian experience becomes increasingly relevant. Australia's retirement system combines a means-tested public pension with mandatory superannuation contributions of 12 percent of income (Australian Taxation Office, n.d.; Swoboda, 2014). While institutional differences remain, the growing role of KiwiSaver makes the two systems more comparable than in the past.

3.1 Spending By Age in New Zealand

The Stats NZ Household Economic Survey (HES) provides the most comprehensive data on New Zealand retirement spending patterns. Two recent reports are particularly relevant. Le and Richardson (2023) analyse expenditure patterns by age across New Zealand households, while Retirement Income Interest Group (2024) focus specifically on spending through retirement and its implications for drawdown strategies.

Using HES data for the year ended June 2019, Le and Richardson (2023) find that expenditure declines sharply at retirement and continues to fall thereafter. Mean household expenditure is \$64,700 for ages 65–74, then 28% lower for ages 75–84, and 44% lower for ages 85+. Expenditure declines across most categories, with the largest falls in mortgage repayments, air travel, and recreation. Unusually, healthcare expenditure also declines with age in New Zealand, likely reflecting the public funding of most core medical services. Retirement Income Interest Group (2024) estimate the real decline at roughly 2% per year after age 65.

The HES data cannot, however, distinguish voluntary from involuntary spending reductions. Retirees may reduce expenditure because they become less active and have genuinely lower needs, or because they face binding financial constraints. Without longitudinal data tracking individuals over time, these explanations cannot be separated cleanly. The observed decline likely reflects both channels.

Understanding the source of this decline is central to evaluating welfare. If spending falls primarily due to financial constraints, optimal strategies should aim to sustain pre-retirement consumption levels. If it reflects voluntary lifestyle adjustment, forcing consumption to remain high may be neither necessary nor welfare-improving. This distinction also affects risk tolerance: financially constrained retirees may need safer portfolios, while those adjusting voluntarily may tolerate more risk.

The retirement spending decline is sometimes called the "retirement consumption puzzle" (Ando & Modigliani, 1963). Standard life-cycle models predict that households should smooth consumption over time: accumulating wealth during working life, then gradually decumulating in retirement while maintaining living standards. Under this framework, retirees should save just enough to sustain pre-retirement consumption, ultimately exhausting wealth near the end of life. Yet empirical evidence shows that many retirees reduce spending at retirement despite having sufficient wealth, and in some cases continue to accumulate wealth during retirement. This suggests either that retirees fail to optimise consumption intertemporally, or that additional factors shape their behaviour.

Whether this pattern reflects genuine welfare loss or merely measurement and composition effects remains unclear. Consumption expenditure is an imperfect proxy for welfare, particularly when higher spending is driven by medical needs or lower spending reflects voluntary lifestyle adjustment. Barrett and Brzozowski (2010) find that households retiring unexpectedly due to adverse health shocks experience sharper spending declines than those retiring as planned, consistent with involuntary constraints. However, much of the observed decline may be voluntary. Retirees spend less on commuting, work clothing, and purchased meals (Aguiar & Hurst, 2008; Battistin et al., 2009), and may substitute market purchases with home production, reducing measured expenditure while maintaining consumption (Becker, 1965; Hurd & Rohwedder, 2003). These mechanisms can mask substantial cross-household heterogeneity in retirement outcomes.

Cross-country comparisons highlight the role of institutional context. Banks et al. (2016) find that U.K. retirees reduce real consumption by 2.2% per year versus 1.4% per year in the U.S. This divergence is driven largely by healthcare financing: U.S. retirees face high out-of-pocket costs that rise with age, while U.K. retirees benefit from comprehensive public coverage. When healthcare spending is excluded, consumption trajectories are more similar across countries.

Australian evidence also shows spending declines of 0.5–1% per year in real terms (Grattan Institute, 2019). Unlike New Zealand's cross-sectional HES data, the Grattan Institute analysis uses longitudinal data tracking individuals over time. The decline persists in both cross-sectional and longitudinal samples, suggesting it reflects genuine behavioural change rather than cohort effects. Importantly, the Grattan Institute finds that most Australian retirees are not financially constrained: fewer than half draw down wealth at all, and over 40% accumulate wealth during retirement. Retirees also report feeling more financially secure than working-age households. As Retirement Income Interest Group (2024) note, this indicates that spending reductions primarily reflect voluntary adjustment to changing lifestyle needs rather than binding constraints.

3.2 Bequests and Precautionary Savings

An aspect of the "retirement consumption puzzle" that has also been studied is why many retirees do not draw down significantly on their wealth during retirement, and in some cases even accumulate more wealth during retirement. This is a well-established phenomenon, having been observed in multiple countries and datasets.

The limited wealth drawdown observed in many countries requires explanation. Hurd (1987) proposed that retirees preserve wealth to leave bequests. However, survey evidence casts doubt on this explanation. Alonso-García et al. (2022) find that Australian and Dutch retirees rarely cite intentional bequest motives beyond provision for a surviving spouse. Instead, retirees prioritise financial independence and quality of life, suggesting that limited decumulation reflects precautionary saving for uncertain future costs.

Precautionary motives are well-supported empirically. De Nardi et al. (2010) show that exposure to uncertain health and long-term care costs explains much of the observed non-decumulation in the United States. Similarly, Banks et al. (2016) find that healthcare costs drive higher late-life spending among U.S. retirees relative to those in the U.K. with broader public coverage. New Zealand's system is mixed: most core medical services are publicly funded, but long-term care such as assisted living and memory care can involve substantial out-of-pocket costs (New Zealand Government, n.d.). Retirement Income Interest Group (2024) conclude that precautionary motives in New Zealand are likely weaker than in the U.S. but

3.3 Spending by Household Composition and Wealth

Whether a retiree lives alone or with a partner is an important determinant of spending patterns, as is home-ownership status. We review this evidence in the next draft iteration once the remaining New Zealand-specific references for this subsection are finalised.

4 Modelling Perceptions of Risk and Utility

Traditional portfolio optimisation (arguably beginning with Markowitz (1952)) typically treats return variance as the risk measure. This follows from the assumption that investors are risk-averse and prefer lower variance for a given expected return. However, for lifetime allocation problems, variance can be difficult to interpret and communicate. In particular, return variance does not directly capture the event investors often care most about: failing to meet spending needs in retirement. Rather, risk of ruin emerges from the interaction between return volatility and withdrawal requirements. For these reasons, variance-based optimisation may yield recommendations that are difficult to interpret and potentially misaligned with investor priorities.

This approach persists in the recent literature. For example, Mäkinen and Toivanen (2024) demonstrate their method using variance and semi-variance of terminal wealth as optimisation objectives for a control-matrix policy. While likely intended as a proof of concept, this objective choice can generate counterintuitive outcomes: the optimiser may truncate the upper tail of terminal wealth in ways that appear misaligned with practical investor concerns. A detailed discussion and replication are provided in the appendix.

Ultimately, whatever risk measure we choose should make sense from a behavioural perspective and, ideally, should be understandable to investors. As discussed in the preceding section, risk perceptions are closely tied to spending needs and concerns about meeting those needs over time. We therefore use that evidence to motivate a risk measure that is interpretable, communicable, and relevant to typical New Zealand retirees.

An essential component of optimising asset allocation strategies is the choice of risk measure, which serves as the objective function to be minimised (or maximised). This choice can materially influence the resulting strategy, because different measures capture different dimensions of risk and therefore imply different return-risk trade-offs.

Our risk measure should be interpretable and directly relevant to outcomes that real-world investors care about. Depending on the optimisation method, it may also need to be differentiable with respect to policy parameters, so that gradient-based methods can be used efficiently.

5 Methods for Simulating Lifetime Wealth Trajectories

A key component of studying lifetime asset allocation is the ability to simulate realistic wealth trajectories for individual investors. This requires modelling both accumulation and decumulation, while accounting for contributions, withdrawals, returns, and spending needs. Our approach uses a Monte Carlo framework that generates a wide range of possible outcomes under alternative allocation policies and market conditions. Importantly, the simulation framework is agnostic to return source: we can use historical, bootstrapped, or model-simulated returns. This flexibility allows us to test strategy robustness across a broad set of assumptions. The framework also accommodates time-varying deposits and withdrawals, which is essential for representing real-world household behaviour.

Unlike the approach proposed by Merton (1969), ours is a discrete-time simulation that models the investor's wealth at regular intervals. This allows us to capture savings and spending behaviour in fixed-dollar amounts, which is more realistic for most retirees.

Each step of the model takes the following form:

$$W_{t+1} = g_{t+1} W_t + \pi_{t+1} \tag{1}$$

Where W_t is the wealth at time t, g_{t+1} is the growth factor on the portfolio between time t and $t+1$, and π_{t+1} is the net contribution (or withdrawal if negative) made at time $t+1$. This can represent regular contributions during the accumulation phase or withdrawals to cover spending needs during the decumulation phase. The model can be easily adapted to include varying contribution and withdrawal amounts over time, allowing for a more nuanced representation of investor behaviour, including simulating variable or stochastic spending patterns.

The growth factor g_{t+1} is determined by the asset allocation strategy being tested, which specifies the proportion of wealth invested in different asset classes. Specifically, we define asset allocation as a vector $\mathbf{a}_t$

at time t, where each element $a_{i,t}$ represents the proportion of wealth invested in asset class i. The portfolio growth factor is then calculated as:

$$g_{t+1} = 1 + \mathbf{a}_t^\top \mathbf{r}_{t+1} \tag{2}$$

Where $\mathbf{r}_{t+1}$ is a vector of returns for each asset class between time t and $t+1$.

In a Monte Carlo simulation, we generate a large number of possible return paths. This means that for each time step, we have N different return vectors for our k asset classes. For each path, we simulate the investor's wealth trajectory over time using the equations above. This means that equation 1 has the matrix form:

$$\mathbf{w}_{t+1} = \mathbf{w}_t \odot \mathbf{g}_{t+1} + \boldsymbol{\pi}_{t+1} \tag{3}$$

Where $\mathbf{w}_t$, $\mathbf{g}_{t+1}$, and $\boldsymbol{\pi}_{t+1}$ are vectors of length N representing the wealth, portfolio returns, and contributions or withdrawals for each of the N simulated paths at time t, and $\odot$ denotes element-wise (Hadamard) multiplication.

The growth factors $\mathbf{g}_{t+1}$ are determined by the asset allocation strategy and a $N \times k$ matrix of returns $\mathbf{R}_{t+1}$ for the k asset classes across the N simulated paths. The asset allocation strategy can be defined in various ways, the first of which is as a function of time (or age). In this case, we define the asset allocation at time t as $\mathbf{a}_t = f(t; \boldsymbol{\theta})$, where f is a function parameterised by a set of parameters $\boldsymbol{\theta}$.

This yields the portfolio growth factors as:

$$\mathbf{g}_{t+1} = 1 + \mathbf{R}_{t+1} \cdot f(t+1; \boldsymbol{\theta}) \tag{4}$$

In this form, the model can accommodate a wide range of asset allocation strategies based on the investor's age. This is similar to the approach that Anarkulova et al. (2025) use, and our first objective will be to replicate their findings for the optimal glide-paths, and to examine how these findings change when we modify the risk measures and spending assumptions to better reflect real-world investor behaviour, particularly in the New Zealand context.

Implications for Consilium and an Aside on Using Matrix Operations and Computational Efficiency

The effort taken to express the model in matrix form is not only for mathematical clarity; it also allows us to leverage vectorised operations through highly optimised linear algebra libraries (for example, `numpy` (Harris et al., 2020)), which can substantially accelerate simulation when path counts and horizon lengths are large. In prior internal work on related Monte Carlo workflows, non-vectorised implementations relying heavily on for-loops have been used, which are computationally inefficient. By contrast, matrix operations allow array-level computation and can produce order-of-magnitude speed improvements. Applied in practice, this may materially improve analysis throughput.

The use of matrix operations will also make it easier when it comes to optimising the asset allocation strategy, as we can leverage automatic differentiation tools (e.g. `tensorflow` (Martín Abadi et al., 2015), `pytorch` (Paszke et al., 2019)) to compute gradients of the risk measure with respect to the parameters of the asset allocation strategy.

5.1 Modelling Dynamic Investor Behaviour

The model described above is flexible enough to accommodate a wide range of asset allocation strategies, but it also allows for the modelling of dynamic investor behaviour in terms of contributions and withdrawals. This is an important feature, as it allows us to simulate more realistic spending patterns that can vary over time and in response to changes in the investor's wealth.

5.1.1 Wealth-Responsive Spending Policy

Rather than having a fixed spending pattern, we can also model the investor's spending as a function of their current wealth level. In some cases, retirees may have a fixed spending floor that they need to meet each year, regardless of the performance of their portfolio. In other cases, retirees may adjust their spending based on the performance of their portfolio, for example by reducing discretionary spending during periods of poor returns. To model this behaviour, we can define the net contribution (or withdrawal) at time t as a function of wealth: $\pi_t = g(W_t; \boldsymbol{\phi})$, where g is a function parameterised by a set of parameters $\boldsymbol{\phi}$. This allows us to simulate more realistic spending patterns that can vary over time and in response to changes in the investor's wealth. For example, we could model a spending policy that has a fixed floor and a variable discretionary component that adjusts based on the investor's current wealth level. This could be represented as:

$$\pi_t = -\max(S_{\text{floor}}, \alpha W_t) \tag{5}$$

Where S_{floor} is the fixed spending floor, and α is a parameter that determines how much of the investor's wealth is allocated to discretionary spending.

Additionally, we can prevent the investor from withdrawing more than their current wealth by defining the net contribution as:

$$\pi_t = -\min(g_{t+1}W_t, \max(S_{\text{floor}}, \alpha W_t)) \tag{6}$$

Equally, we could model a spending policy that has a random component, for example by adding a stochastic term to the spending function modelling rare, but significant, unexpected expenses that retirees may face, such as healthcare costs or home repairs.

5.1.2 Wealth-Responsive Allocation Policy

Like with the spending policy, we can also model the asset allocation strategy as a function of both time and wealth. In the age-based glide-path framework, the hypothetical investor's asset allocation strategy is only responsive to time (i.e. age), and does not take into account the investor's current wealth level. In practice, an investor's risk tolerance and spending needs may vary significantly depending on their accumulated wealth at any given time. For example, an investor with a large portfolio may be able to tolerate more risk than an investor with a smaller portfolio, even if they are the same age. To address this limitation, we will attempt to extend the framework using a method similar to that proposed by Mäkinen and Toivanen (2024), who introduce a control matrix that allows the asset allocation strategy to depend on both time and wealth.

In this framework, the asset allocation at time t and wealth W_t is given by $\mathbf{a}_t = C(t, W_t; \boldsymbol{\theta})$, where C is the control matrix parameterised by $\boldsymbol{\theta}$. This yields the portfolio growth factors as:

$$\mathbf{g}_{t+1} = 1 + \mathbf{R}_{t+1} \cdot C(t+1, W_t; \boldsymbol{\theta}) \tag{7}$$

The way that this control-matrix works is that it defines a grid over the time and wealth dimensions, with each cell in the grid corresponding to a specific asset allocation. During the simulation, the investor's current age and wealth level are used to determine which cell in the grid they fall into, and thus the corresponding allocation to the risky asset (the remainder of the portfolio is allocated to the safe asset).

This approach allows for a more nuanced asset allocation strategy that can adapt to the investor's changing circumstances over time. For example, an investor who has accumulated significant wealth may be able to take on more risk, while an investor with lower wealth may need to adopt a more conservative approach. However, it also exponentially increases the number of parameters that need to be optimised, which can make any optimisation process more challenging.

However, we can reduce the search-space somewhat by using less granular grid points for the control matrix and interpolating between them to determine the asset allocation for time steps and wealth levels that fall between the defined grid points. This allows us to maintain a reasonable level of flexibility in the asset allocation strategy while keeping the number of parameters manageable. Moreover, it is unlikely that investors' asset allocation policy will be meaningfully different for small increments in time or wealth, so using a coarser grid with interpolation is unlikely to significantly reduce the optimality of the resulting strategy, while making the optimisation process more tractable. This idea of interpolating between grid points was first proposed by Mäkinen and Toivanen (2024), who use bilinear interpolation to determine the asset allocation for wealth levels that fall between the defined wealth-grid points, however, we also propose that interpolation could be used for time steps that fall between the defined time-grid points as well, which would further reduce the number of parameters that need to be optimised.

To illustrate this more concretely, consider Figure 2, which shows a simple control matrix for a two-asset portfolio. The matrix has six time steps (0 to 5) and five wealth levels (0 to 4), giving a 6 by 5 grid. Each cell corresponds to a risky-asset allocation (with the remainder allocated to the safe asset). For instance, if an investor is at time step 2 with wealth level 3, the policy returns the allocation in that cell. For time or wealth values between grid points, we use the interpolation method proposed by Mäkinen and Toivanen (2024). For example, at time 2.5 and wealth 3.5, bilinear interpolation uses the four neighbouring grid points to determine allocation.

For allocations with more than two asset classes, the control matrix can be extended to a higher-dimensional tensor, with each additional dimension representing an additional asset class. Mäkinen and Toivanen (2024)

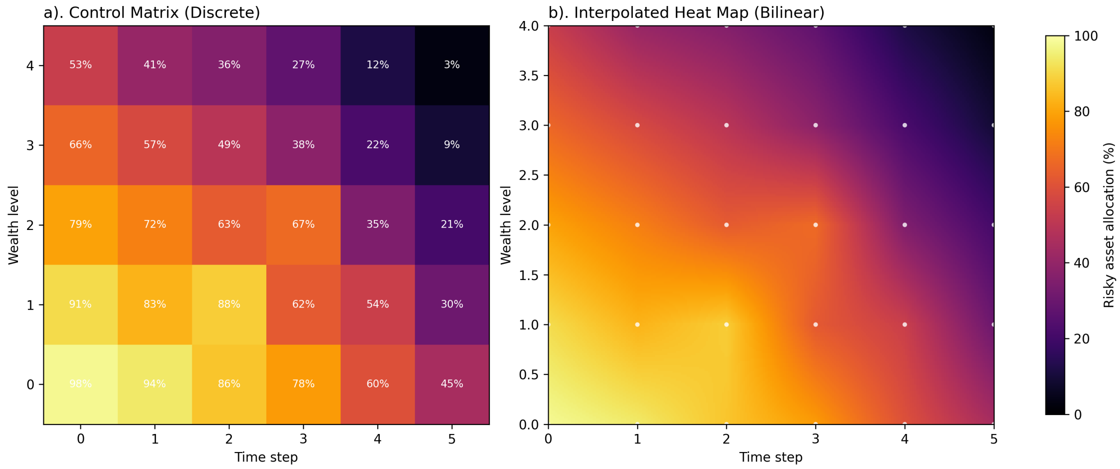

Figure 2: Illustrative example of a discrete control matrix (a) and bilinearly interpolated allocation surface (b).

use bilinear interpolation to determine the asset allocation for wealth levels that fall between the defined grid points, ensuring a smooth transition between different allocation strategies.

5.2 Methods for Simulating Asset Returns

6 Methods for Optimising Asset Allocation Strategies

The key challenge in optimising an asset allocation strategy is that the parameter space can be extremely large, even for simple policy classes. For example, for a two-asset age-based glide-path with allocations in 10 percent increments over 60 years, there are 11 possible allocations at each time step, giving 11^{60} possible strategies. This search space is intractable, so numerical optimisation is required.

Merton (1969) use a formulation in which the investor maximises a lifetime utility function over consumption. The framework is mathematically tractable, but it relies on proportional consumption from wealth each period, which is often unrealistic for retirees facing fixed-dollar spending needs. In addition, the closed-form solution assumes IID returns, precisely the return structure that Anarkulova et al. (2025) argue can bias conclusions about declining-risk optimality. For our purposes, this framework is therefore limited.

Anarkulova et al. (2025) use a recursive approach in which grid search is used to update allocation at each time step, with forward passes repeated until convergence. This approach is likely suitable for age-based glide-path optimisation, but extension to a time-by-wealth control matrix is less straightforward because the parameter count increases by roughly an additional order of magnitude.

In their paper, Mäkinen and Toivanen (2024) optimise a control-matrix allocation strategy using gradient-based methods and an adjoint formulation, which is computationally appealing for high-dimensional policy spaces. This provides a practical starting point for our own optimisation design, while still allowing us to adapt the objective function to better reflect retiree-relevant risk.

7 Optimal Lifetime Asset Allocation Strategies for New Zealand Retirees

Having established the simulation and optimisation framework, we now apply it to evaluate candidate lifetime allocation policies in the New Zealand retirement context.

7.1 Optimal Age-Based Glide-Paths

7.2 Optimal Wealth-Responsive Allocation Strategies

8 Discussion and Implications

9 Conclusion

References

Aguiar, M., & Hurst, E. (2008). *Deconstructing lifecycle expenditure* (tech. rep. No. 13893). National Bureau of Economic Research. https://doi.org/10.3386/w13893

Alonso-García, J., Bateman, H., Bonekamp, J., van Soest, A., & Stevens, R. (2022). Saving preferences after retirement. *Journal of Economic Behavior & Organization*, *198*, 409–433. https://doi.org/10.1016/j.jebo.2022.04.005

Anarkulova, A., Cederburg, S., & O'Doherty, M. S. (2025, March). Beyond the status quo: A critical assessment of lifecycle investment advice [SSRN Working Paper]. https://doi.org/10.2139/ssrn.4590406

Ando, A., & Modigliani, F. (1963). The "life cycle" hypothesis of saving: Aggregate implications and tests. *The American Economic Review*, *53*(1), 55–84. https://www.jstor.org/stable/1817129

Australian Taxation Office. (n.d.). *Super guarantee percentage* [Retrieved 6 July 2023].

Banks, J., Blundell, R., Levell, P., & Smith, J. P. (2016). *Life-cycle consumption patterns at older ages in the us and the uk: Can medical expenditures explain the difference?* (Tech. rep. No. 22513). National Bureau of Economic Research. https://doi.org/10.3386/w22513

Barrett, G., & Brzozowski, M. (2010). *Involuntary retirement and the resolution of the retirement-consumption puzzle: Evidence in australia* (tech. rep.). University of Sydney, School of Economics.

Battistin, E., Brugiavini, A., Rettore, E., & Weber, G. (2009). The retirement consumption puzzle: Evidence from a regression discontinuity approach. *American Economic Review*, *99*(5), 2209–2226. https://doi.org/10.1257/aer.99.5.2209

Becker, G. S. (1965). A theory of the allocation of time. *The Economic Journal*, *75*(299), 493–517. https://doi.org/10.2307/2228949

Bengen, W. P. (1996). Asset allocation for a lifetime. *Journal of Financial Planning*, *9*(4), 86–92. https://www.financialplanningassociation.org/article/journal/AUG96-asset-allocation-lifetime

Bodie, Z., Kane, A., & Marcus, A. J. (2023). *Investments*. McGraw-Hill US Higher Ed ISE. https://ebookcentral.proquest.com/lib/canterbury/detail.action?docID=7190049

Bodie, Z., Merton, R. C., & Samuelson, W. F. (1992). Labor supply flexibility and portfolio choice in a life cycle model. *Journal of Economic Dynamics and Control*, *16*(3), 427–449. https://doi.org/https://doi.org/10.1016/0165-1889(92)90044-F

Campbell, J. Y. (1996). Understanding risk and return. *Journal of Political Economy*, *104*(2), 298–345.

Choi, J. J. (2022). Popular personal financial advice versus the professors. *Journal of Economic Perspectives*, *36*(4), 167–192. https://www.jstor.org/stable/27171135

Cocco, J. F., Gomes, F. J., & Maenhout, P. J. (2005). Consumption and portfolio choice over the life cycle. *The Review of Financial Studies*, *18*(2), 491–533. Retrieved December 8, 2025, from http://www.jstor.org/stable/3598044

De Nardi, M., French, E., & Jones, J. B. (2010). *Differential mortality, uncertain medical expenses, and the saving of elderly singles* (tech. rep. No. 16453). National Bureau of Economic Research. https://doi.org/10.3386/w16453

Dexter, G. (2025, November 24). *Kiwisaver provider calls for increased contribution to be compulsory*. Retrieved December 6, 2025, from https://www.rnz.co.nz/news/political/579777/kiwisaver-provider-calls-for-increased-contribution-to-be-compulsory

Dolvin, S. D., Templeton, W. K., & Rieber, W. (2010). Asset allocation for retirement: Simple heuristics and target-date funds. *Scholarship and Professional Work - Business*, (17). https://digitalcommons.butler.edu/cob_papers/17

Estrada, J. (2014). The glidepath illusion: An international perspective. *The Journal of Portfolio Management*, *40*(4), 52–64. https://doi.org/10.3905/jpm.2014.40.4.052

Estrada, J. (2015). The retirement glidepath: An international perspective. *The Journal of Investing*, *25*(2), 28–36. https://doi.org/10.3905/joi.2016.25.2.028

Financial Markets Authority. (2023, August 9). *Kiwisaver default funds*. Retrieved December 6, 2025, from https://www.fma.govt.nz/consumer/kiwisaver-and-superannuation/about-kiwisaver/kiwisaver-default-funds/

Financial Markets Authority. (2024, June 24). *Spotlight on: Balanced funds*. Retrieved December 6, 2025, from https://www.fma.govt.nz/library/articles/spotlight-on-balanced-funds/

Ghysels, E., Santa-Clara, P., & Valkanov, R. (2005). There is a risk-return trade-off after all. *Journal of Financial Economics*, *76*(3), 509.

Gilbert, A. (2025, November 27). *Lifting kiwisaver contributions to 12% makes sense – when the whole scheme is fixed*. Retrieved December 6, 2025, from https://www.nzherald.co.nz/nz/lifting-kiwisaver-contributions-to-12-makes-sense-when-the-whole-scheme-is-fixed

Grattan Institute. (2019). *The savings behaviour of older households: Submission 435 – supplementary submission to inquiry into the implications of removing refundable franking credits* (Supplementary submission No. 435) (Draws from: Daley, J., Coates, B., Wiltshire, T., Emslie, O., Nolan, J., and Chen, T. (2018), "Money in Retirement: More than Enough", Grattan Institute). Parliament of Australia.

Harris, C. R., Millman, K. J., van der Walt, S. J., Gommers, R., Virtanen, P., Cournapeau, D., Wieser, E., Taylor, J., Berg, S., Smith, N. J., Kern, R., Picus, M., Hoyer, S., van Kerkwijk, M. H., Brett, M., Haldane, A., del Río, J. F., Wiebe, M., Peterson, P., … Oliphant, T. E. (2020). Array programming with NumPy. *Nature*, *585*(7825), 357–362. https://doi.org/10.1038/s41586-020-2649-2

Hurd, M. D. (1987). Savings of the elderly and desired bequests. *The American Economic Review*, *77*(3), 298–312. https://www.jstor.org/stable/1804093

Hurd, M. D., & Rohwedder, S. (2003). *The retirement-consumption puzzle: Anticipated and actual declines in spending at retirement* (tech. rep. No. 9586). National Bureau of Economic Research. https://doi.org/10.3386/w9586

Ibbotson, R., & Kaplan, P. (2001). Does asset allocation policy explain 40, 90, 100 percent of performance? *Yale School of Management, Yale School of Management Working Papers*, *56*. https://doi.org/10.2469/faj.v56.n1.2327

Le, T., & Richardson, E. (2023, August). *Expenditure patterns of new zealand retiree households* (Motu Working Paper No. 23-07). Motu Economic and Public Policy Research. https://motu.nz/our-research/population-and-labour/individual-and-group-outcomes/expenditure-patterns-nz-retiree-households

Lundblad, C. (2007). The risk return tradeoff in the long run: 1836–2003. *Journal of Financial Economics*, *85*(1), 123–150. https://doi.org/10.1016/j.jfineco.2006.06.003

Luxon, C. (2025, November 23). *National to lift kiwisaver contributions*. Retrieved December 6, 2025, from https://www.national.org.nz/news/20251123-hon-christopher-luxon-national-to-lift-kiwisaver-contributions

Mäkinen, R. A. E., & Toivanen, J. (2024). Monte carlo expected wealth and risk measure trade-off portfolio optimization. *SIAM Journal on Financial Mathematics*.

Markowitz, H. (1952). Portfolio selection. *The Journal of Finance*, *7*(1), 77–91. Retrieved August 6, 2025, from http://www.jstor.org/stable/2975974

Martín Abadi, Ashish Agarwal, Paul Barham, Eugene Brevdo, Zhifeng Chen, Craig Citro, Greg S. Corrado, Andy Davis, Jeffrey Dean, Matthieu Devin, Sanjay Ghemawat, Ian Goodfellow, Andrew Harp, Geoffrey Irving, Michael Isard, Jia, Y., Rafal Jozefowicz, Lukasz Kaiser, Manjunath Kudlur, … Xiaoqiang Zheng. (2015). TensorFlow: Large-scale machine learning on heterogeneous systems [Software available from tensorflow.org]. https://www.tensorflow.org/

Merton, R. C. (1969). Lifetime portfolio selection under uncertainty: The continuous-time case. *The Review of Economics and Statistics*, *51*(3), 247–257. https://doi.org/10.2307/1926560

Ministry of Business, Innovation and Employment and The Treasury. (2019, August 7). *Review of kiwisaver default provider arrangements* [Consultation document]. Retrieved December 6, 2025, from https://www.treasury.govt.nz/publications/consultation/review-kiwisaver-default-provider-arrangements

Ministry of Social Development. (2022). *Description of new zealand's current retirement income policies* (tech. rep.) (Background paper prepared for the Retirement Commissioner's 2022 Review of Retirement Income Policy). Ministry of Social Development. Retrieved March 17, 2026, from https://assets.retirement.govt.nz/public/Uploads/Retirement-Income-Policy-Review/2022-RRIP/Description-of-New-Zealands-Current-Retirement-Income-Policies-MSD-for-2022-RRIP-.pdf

New Zealand Government. (n.d.). *Paying for residential care*. Retrieved April 13, 2026, from https://www.govt.nz/browse/health/rest-homes-and-residential-care/pay-for-residential-care/

OECD. (2025). *Pension markets in focus 2025*. OECD Publishing. Paris. https://doi.org/10.1787/b095d0a0-en

Palmer, R. (2025, September 7). *Watch: Peters promises compulsory 10% kiwisaver and tax cuts*. Retrieved December 6, 2025, from https://www.rnz.co.nz/news/national/572352/watch-peters-promises-compulsory-10-percent-kiwisaver-and-tax-cuts

Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Killeen, T., Lin, Z., Gimelshein, N., Antiga, L., Desmaison, A., Köpf, A., Yang, E., DeVito, Z., Raison, M., Tejani, A., Chilamkurthy, S., Steiner, B., Fang, L., … Chintala, S. (2019). Pytorch: An imperative style, high-performance deep learning library. https://arxiv.org/abs/1912.01703

Retirement Commission Te Ara Ahunga Ora. (n.d.). *A snapshot of the history of kiwisaver* [PDF pamphlet]. Retrieved December 6, 2025, from https://assets.retirement.govt.nz/public/Uploads/Retirement-Income-Policy-Review/TAAO-A-Snapshot-of-the-History-of-KiwiSaver.pdf

Retirement Commission Te Ara Ahunga Ora. (2022, August 15). *New research highlights how reliant older new zealanders are on nz super*. Retrieved March 17, 2026, from https://retirement.govt.nz/news/latest-news/new-research-highlights-how-reliant-older-new-zealanders-are-on-nz-super

Retirement Income Interest Group. (2024, December). *Spending patterns through retirement: Implications for retirement planning and drawdown* (Report). New Zealand Society of Actuaries. https://actuaries.org.nz/content/uploads/2024/12/Spending-through-retirement-RIIG-Dec2024.pdf

Shiller, R. (2005). Life-cycle portfolios as government policy. *The Economists' Voice*, *2*(1), 14–14. https://doi.org/10.2202/1553-3832.1095

Shiller, R. J. (2006). Life-cycle personal accounts proposal for social security: An evaluation of president bush's proposal. *Journal of Policy Modeling*, *28*(4), 427–444. https://doi.org/https://doi.org/10.1016/j.jpolmod.2005.10.010

SuperLife. (2026, March 26). *Statement of investment policy and objectives (sipo): Superlife invest*. Retrieved March 31, 2026, from https://www.superlife.co.nz/files/LegalDocs/SIPO/SuperLife_Invest_-_Statement_of_Investment_Policy_and_Objectives.pdf

Swoboda, K. (2014, March 11). *Major superannuation and retirement income changes in australia: A chronology* (Research Papers 2013–14). Australian Parliamentary Library.

Te Ara Ahunga Ora: Retirement Commission. (2021). *Review of retirement income: Policy paper 03* (Policy Paper No. 03). Te Ara Ahunga Ora: Retirement Commission. https://assets.retirement.govt.nz/public/Uploads/Retirement-Income-Policy-Review/TAAO-RC-Policy-Paper-2021-03.pdf

The Treasury. (2025, September 24). *He tirohanga mokopuna 2025: Statement on the long term fiscal position* (Retrieved from https://www.treasury.govt.nz/publications/ltfp/he-tirohanga-mokopuna-2025 on December 6, 2025). The Treasury. New Zealand Government. Retrieved December 6, 2025, from https://www.treasury.govt.nz/publications/ltfp/he-tirohanga-mokopuna-2025

A Replication and Discussion of Control Matrix Optimisation as Proposed by Mäkinen and Toivanen (2024)

Mäkinen and Toivanen (2024) formulate their Monte Carlo optimisation in a way that is similar to the framework described in the main text. As in our setup, they use a control matrix that allows asset allocation to respond jointly to age and wealth. Their objective function is based on terminal wealth moments, specifically variance or semi-variance. They then compute gradients of this objective with respect to control-matrix parameters using adjoint methods, which enables efficient gradient-based optimisation (in their case, a quasi-Newton method with BFGS updates).

Because this method is mathematically elegant and computationally efficient, we have attempted to replicate it here. The code for this replication can be found in the accompanying repository, and the results are shown in Figure 3 for the mean-variance and mean-semi-variance cases.

In line with the setup described by Mäkinen and Toivanen (2024), the replication uses a Monte Carlo horizon of $T = 20$ years with time step $\Delta t = 0.25$, giving $N = 80$ time steps. The policy is represented on a time-wealth control grid with $M = 31$ wealth nodes, upper wealth bound $W_{\max} = 30$, and bilinear interpolation between nodes. In the notebook, wealth is clipped to $[0, W_{\max}]$ before interpolation. Optimisation is performed with L-BFGS-B (a quasi-Newton method) with box constraints.

For the one-risky-asset, no-leverage case used in Figure 3, the full parameterisation is summarised in Table 1.

Table 1: Parameter setup for the one-risky-asset mean-variance replication.

Parameter	Value
Risk-free rate (r)	0.03
Risky-asset drift (μ_1)	0.0795
Risky-asset volatility (σ_1)	0.15
Contribution rate (π)	0.1
Initial wealth (w_0)	1
Minimum risky allocation ($p_{\min}$)	0
Maximum risky allocation ($p_{\max}$)	1
Number of simulation paths (K)	10^5
Risk-aversion weight (λ)	0.1

The replication results are broadly consistent with those reported by Mäkinen and Toivanen (2024). However, they also highlight important limitations. First, as an objective for a retirement investor, terminal-wealth variance is not especially intuitive. In our replication, the terminal-wealth distribution becomes bimodal because the optimiser shifts to a highly conservative policy at high wealth levels, truncating the upper tail that would normally appear under a log-normal-like process. Although this behaviour is mathematically consistent with variance minimisation, it is not obviously aligned with investor welfare. A similar pattern appears under semi-variance, albeit at higher wealth levels. Because semi-variance is computed relative to the mean, higher-return strategies can increase the penalty by raising the mean itself, again encouraging upper-tail truncation. This reinforces the need for risk measures that are both interpretable and behaviourally relevant.

Additionally, the control matrix has limited effective resolution in early periods: many simulated paths map to only a few low-wealth nodes because initial wealth is far below the fixed upper wealth bound. This reduces the optimiser's ability to distinguish trajectories precisely when policy differences likely matter most due to compounding. It also creates many redundant parameters at wealth levels that are never reached (upper-left region in Figure 3). For this reason, we propose calibrating the control-matrix upper bound to the simulated wealth distribution so that resolution is concentrated where paths actually lie.

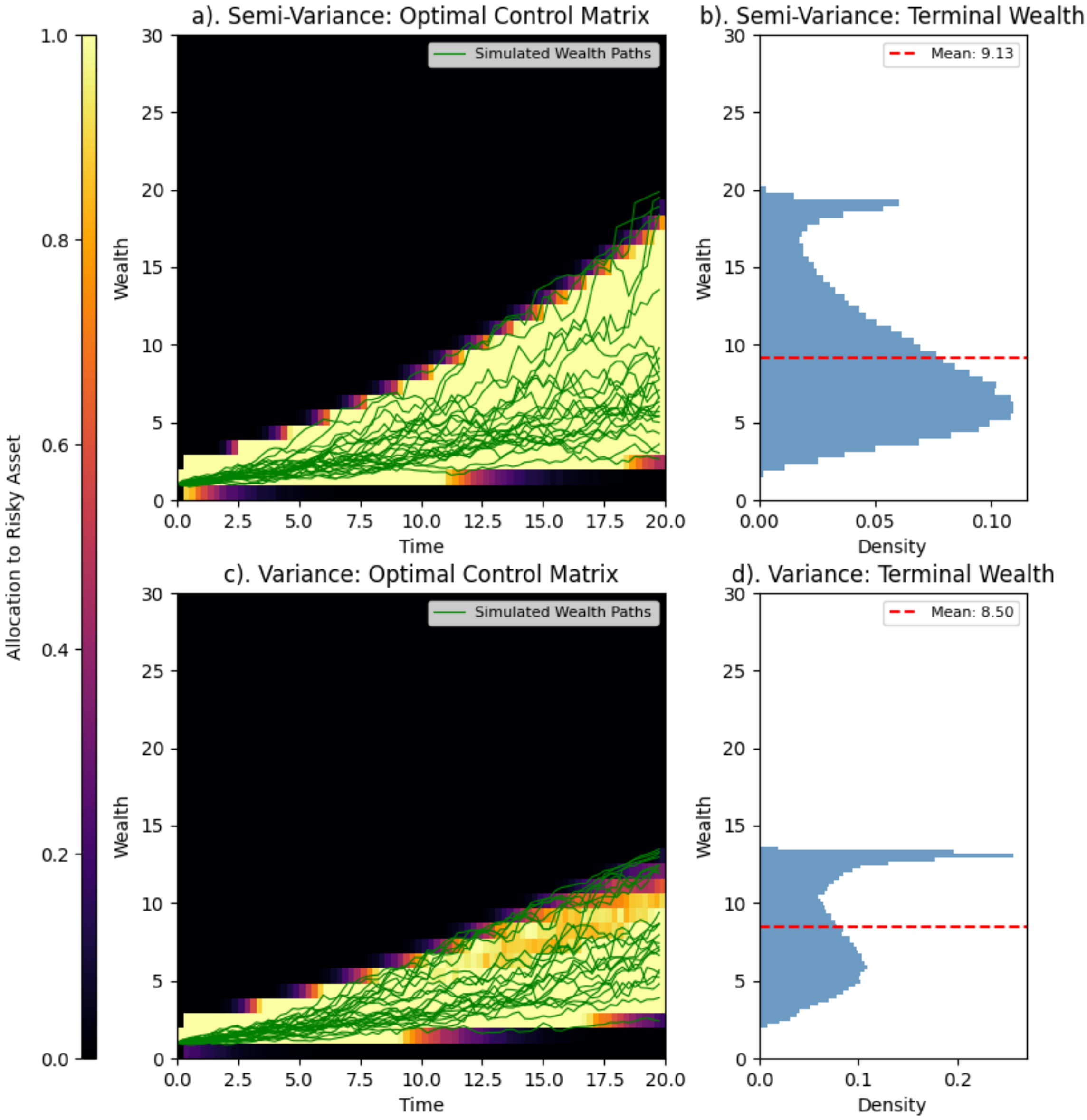

Figure 3: Replication of the control matrix optimization method proposed by Mäkinen and Toivanen (2024) for one risky asset and no leverage. The left column shows the optimal control matrix for the mean-semi-variance case (top) and the mean-variance case (bottom). The right column shows the corresponding histograms of terminal wealth for each risk measure respectively.